

When "Simple" Becomes "Hard": Examining the Fragility of Unplugged AI Toolkits for Abstract Logic Design

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Figure 1: Divergent interaction modes. The same unplugged toolkit produced different patterns based on context. (Left) Urban teachers engage in iterative co-design (standing, physical manipulation). (Right) Rural teachers exhibit "form-filling" behaviors (seated, individual transcription) due to the abstraction gap.

Abstract

Unplugged AI toolkits are promoted as accessible entry points for AI literacy within HCI and AI education. However, this assumption remains underexplored for systems whose core logic lacks direct sensory analogues. We report a comparative field study of two card-based toolkits—one addressing deepfakes, the other LLM-based agents—deployed in workshops with over 140 K-12 teachers across urban and rural contexts. The deepfake toolkit consistently supported engagement and reasoning across all settings. The agent toolkit, however, elicited divergent patterns: in urban workshops, peer scaffolding from participants with partial generative AI experience enabled productive co-design, while in rural workshops lacking such experience, groups frequently abandoned the cards or bypassed the design process altogether. A facilitator-initiated live software demonstration prompted a notable shift, with groups retrieving previously discarded cards and revising their designs.

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These observations suggest that the effectiveness of unplugged representations is not inherent but depends on whether learners possess experiential referents that make the toolkit's abstractions interpretable. We discuss boundary conditions for unplugged approaches and consider the potential value of sequencing experiential encounters before card-based design activities.

CCS Concepts

• **Human-centered computing** → HCI theory, concepts and models; Empirical studies in HCI.

Keywords

AI Literacy, Unplugged Toolkit, Generative AI, Mental Model, K-12 Education

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1 Introduction

As generative AI systems increasingly shape everyday life, AI literacy has become a critical educational priority [9, 12]. Beyond technical skills, AI literacy involves reasoning about how AI systems operate, where their limitations lie, and how they should be used responsibly [9, 10]. For non-technical learners, however, making AI mechanisms intelligible remains challenging—especially when core operations are abstract and difficult to observe [1, 4]. Within HCI and AI education, unplugged and tangible approaches are widely promoted as accessible, low-barrier entry points, particularly in low-resource settings [6, 8]. By externalizing computation into physical artifacts—such as cards, blocks, or canvases—unplugged designs aim to lower technical barriers, reduce reliance on digital infrastructure, and support inclusive participation [5, 8]. This rationale has been especially influential in low-resource and rural education initiatives, where unplugged tools are often positioned as inherently simpler, more accessible, and more equitable alternatives to software-based instruction [11].

Yet existing evidence for unplugged AI education largely focuses on systems with perceptually visible effects, such as image manipulation or recommendation outcomes [2, 3]. Far less is known about how unplugged approaches function for abstract AI systems, including LLM-based agents, whose internal logic lacks direct sensory analogues [7, 13]. In such cases, physical simplicity may conceal substantial cognitive demands. To examine this tension, we conducted a comparative field study of two unplugged AI card toolkits deployed in K–12 teacher workshops across urban and rural contexts. Our findings suggest that the effectiveness of unplugged representations depends on whether learners possess experiential referents that make the toolkits’ abstractions interpretable—a condition that held for one toolkit but not the other. Based on these observations, we discuss the boundary conditions under which unplugged approaches support or hinder learning, and reflect on how brief experiential encounters may complement card-based design activities.

2 Toolkit Design and Study Context

We designed two card-based unplugged toolkits that share a common tangible format—function cards, background cards, ethics cards, and a design canvas—but differ in their underlying system logic and the visibility of the concepts they represent (Figure 2). By holding the medium constant while varying the nature of abstraction, we use these toolkits as contrasting conditions for examining how unplugged representations function across contexts.

2.1 Toolkit A: Deepfake Ethics Toolkit

Composition. The toolkit contains 12 function cards decomposing deepfake techniques into individually recognizable media operations (e.g., Face Swapping, Voice Cloning, Emotion Transfer), 6 context cards providing localized application scenarios (e.g., livestream commerce in rural markets, virtual companionship for left-behind children), and 6 ethics cards addressing social risks such as privacy violation, emotional manipulation, and misinformation. A scenario design canvas serves as the shared physical workspace for each group.

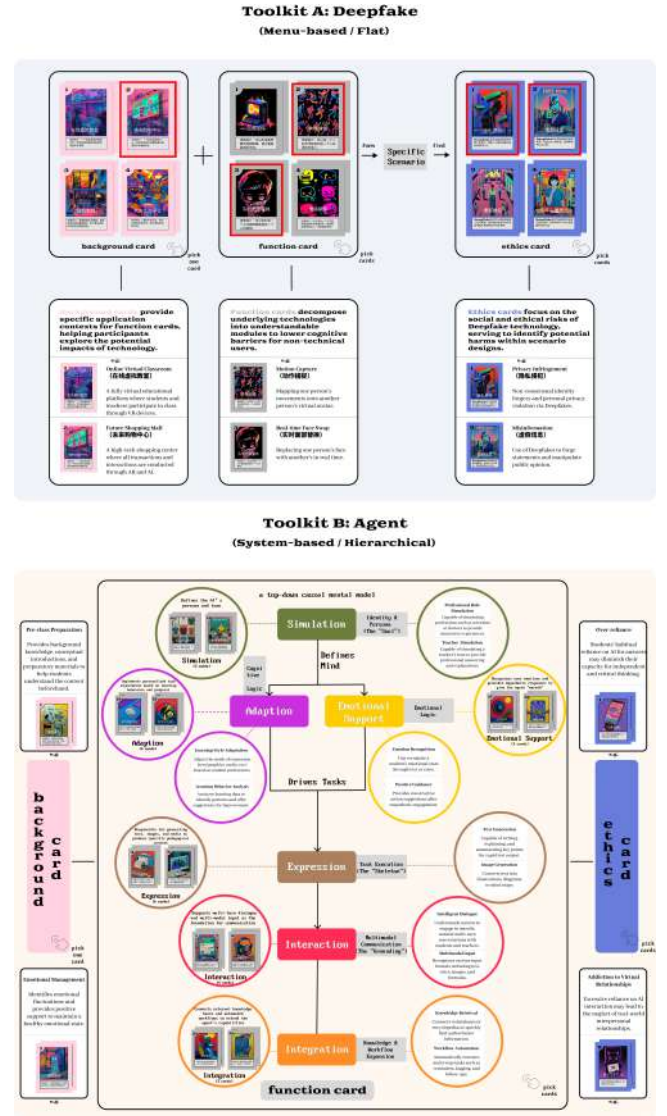


Figure 2: Contrasting System Logic. The Deepfake Toolkit (Top) employs a flat, menu-based structure mapping directly to visible outcomes. In contrast, the Agent Toolkit (Bottom) requires a hierarchical, system-based construction, requires a top-down causal mental model, moving from identity to functional manifestation.

Interaction flow. The toolkit follows a three-phase spiral workflow (Figure 3, Top). In the first phase (unconstrained assembly), groups select one context card and several function cards to freely compose a deepfake application scenario—without ethical constraints—simulating unreflective technology adoption. In the second phase (cross-group ethical review), groups exchange canvases and use ethics cards to stress-test each other’s designs, identifying potential harms. In the third phase (positive redesign), groups reconstruct their scenarios into ethically grounded educational applications based on the vulnerabilities surfaced during review.

Each function card maps directly to a visible or easily imaginable media effect, requiring selection and combination rather than causal reasoning about hidden mechanisms. This flat, menu-based structure (Figure 2, top) allows participants to engage immediately by treating cards as labels for phenomena they have already encountered.

2.2 Toolkit B: AI Agent Toolkit

Composition. The toolkit contains 6 background cards defining K-12 educational challenges (e.g., differentiated instruction, motivation and emotional support), 21 function cards grouped into six progressively abstract agent capabilities: Interaction (4), Expression (6), Adaptation (4), Affective Support (3), Simulation (3), and Integration (3), and 11 ethics cards based on a T.E.A.C.H. framework (Transparency, Equity, Autonomy, Care, Honesty). A unified design canvas structures the full workflow from problem definition to logic specification.

Interaction flow. The toolkit enforces a constrained, sequential process (Figure 3, Bottom). Groups first select a background card and define their agent's persona—name, personality, and core objective—on the canvas. They then select 3–5 function cards, but each selection must be justified through an explicit causal sentence template: "Because [core challenge], we choose [function card]; specifically, when [user action], the agent will [behavior]." This mechanism forces participants to externalize reasoning about invisible system dependencies rather than simply listing features. Following construction, groups conduct a cross-group ethical review using ethics cards, then converge their design into a single-sentence minimum viable product (MVP) description.

Unlike the Deepfake toolkit's flat structure, the Agent toolkit requires hierarchical, causal reasoning: defining high-level properties before selecting supporting components, and articulating dependencies between them (Figure 2, Bottom). This system-based decomposition places substantially greater demands on participants' ability to construct internal mental models of how agent systems operate.

2.3 Workshop Deployment

Participants and sites. We conducted workshops with over 140 K-12 teachers across urban and rural settings in China, spanning more than 10 schools. Urban participants were based in Beijing; rural workshops took place in Shanxi and Inner Mongolia. Urban teachers had varied but generally higher levels of prior AI exposure, including some familiarity with generative AI tools. Rural teachers had limited or no prior experience with AI systems beyond general smartphone use, including short-video platforms.

Procedure. Deepfake workshops lasted approximately 90 minutes, structured as: introduction with real-case demonstration (15 min), scenario assembly (20–25 min), cross-group ethical review (20 min), positive redesign (20 min), and group presentation (15 min), as shown in Figure 3, top. Agent workshops lasted approximately 180 minutes, structured in three acts: problem exploration and scenario selection (45 min), agent blueprint construction including persona definition and causal logic specification (45 min), and ethical review, MVP convergence, and summary (90 min). Groups of

4–6 teachers worked collaboratively at shared tables, as shown in Figure 3, bottom. During the rural Agent workshops, facilitators introduced a live software demonstration (Coze) as a responsive intervention when groups encountered persistent difficulty; this was not a pre-planned experimental condition but an adaptive facilitation decision, and its effects are reported in Section 3.3.

Data collection. Data included photographs of card-based artifacts and design canvases, field observation notes documenting group interaction dynamics, and participant statements recorded during and after sessions. The study was IRB-approved, and informed consent was obtained from all participants. Analysis focused on identifying recurring interaction patterns, moments of breakdown, and shifts in how participants engaged with the toolkits across contexts.

3 Findings

We present observations from workshops using two unplugged toolkits across urban and rural contexts. Although both toolkits were developed under the same design philosophy, they elicited notably different engagement patterns.

3.1 Deepfake Toolkit: Engagement Across Contexts

In workshops using the Deepfake Toolkit, we observed high levels of engagement across both urban and rural contexts. Rural teachers, despite lacking formal AI training, frequently connected card labels to phenomena they had already encountered on short-video platforms. For example, teachers recognized "Face Swapping" not as a technical term but as something they had seen circulating in everyday media. They treated the cards as names for things they already recognized rather than unfamiliar technical definitions.

Artifacts produced in these sessions frequently featured localized scenarios, such as fraud targeting elderly villagers, with active debate around ethics cards. Across both contexts, teachers engaged in sustained reasoning with the cards without requesting additional technical explanation.

"At least I can pass this [fraud prevention awareness] directly to the kids." — Rural teacher

3.2 AI Agent Toolkit: Divergent Engagement Patterns

The Agent Toolkit elicited markedly different patterns across contexts. In urban settings, despite varied individual knowledge levels, peer scaffolding emerged spontaneously: teachers with partial generative AI familiarity guided teammates through card meanings, resulting in iterative, visually layered co-design artifacts. Groups frequently revisited and rearranged cards as their understanding developed through discussion. In rural settings, however, workshops encountered substantial difficulties. In the Shanxi workshop, 50% of the groups dissolved mid-session, merging with neighboring groups because, as participants reported, they "didn't know how to proceed." Teachers described the card labels as unclear and asked facilitators repeatedly what individual terms meant before attempting any design activity. In Inner Mongolia, groups that did complete the task produced artifacts that appeared as linear lists of

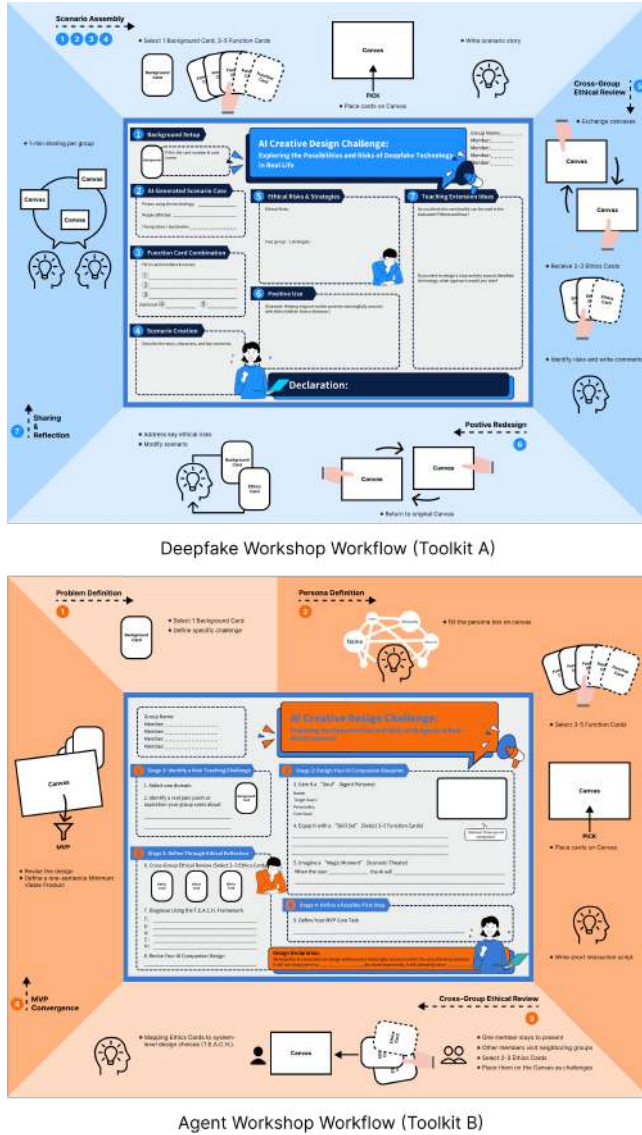


Figure 3: Canvas-guided interaction workflows. Toolkit A (Top): Participants engage in modular assembly and narrative redesign, mapping cards directly to lived experiences. **Toolkit B (Bottom):** The canvas enforces a constrained protocol, requiring participants to define a persona and use causal sentence templates (e.g., "When..., then...") to externalize hidden system logic.

functions—resembling a menu-selection format—rather than the intended relational workflows. Cards were placed onto the canvas sequentially and treated as items to be filled in rather than components to be debated or rearranged.

We also observed a recurring behavior in which participants bypassed the card-based design process. Rather than using the cards to construct workflows, some groups opened AI applications on their phones, posed questions directly to chatbots, and transcribed

the outputs onto the canvas. In one group, a teacher asked a chatbot to "design an educational agent," then copied the response verbatim onto the poster without discussing the individual cards with her teammates. In another, two participants spent the majority of the session comparing chatbot outputs rather than manipulating the physical cards. In these instances, the AI system was treated as a source of ready-made answers rather than an object to be designed.

"What is the use of doing this? If you taught me an AI tool, at least I could use it." — Rural teacher

3.3 Facilitator Intervention: Live Demonstration and Subsequent Shift

A notable shift occurred when facilitators introduced a live software demonstration (**Coze**) as a responsive intervention during the rural Agent Toolkit workshops. The demonstration allowed teachers to observe an AI agent executing tasks in real time, including how different configurations produced different outputs. One teacher who had previously set aside the **"Simulation"** card provided a clear example of this shift. During the demonstration, she observed how changing the System Prompt in Coze generated entirely different responses to the same input question. She then retrieved the card she had earlier discarded, placed it back into her workflow, and revised the surrounding design. She remarked: *"So this is what the role setting really means!"*

This pattern was not isolated. Following the demonstration, several groups retrieved previously set-aside cards and began revising their workflow designs—a behavior not observed prior to the intervention. Group conversations also shifted: whereas pre-demonstration talk centered on asking facilitators what cards meant, post-demonstration discussion increasingly involved participants explaining card functions to each other by referencing what they had just observed in the software.

4 Discussion

Our findings suggest that the effectiveness of unplugged representations is not inherent but situated: it depends on whether learners possess experiential referents that allow them to interpret the toolkit's abstractions meaningfully [6, 7]. Rather than offering a general critique of unplugged approaches, we reflect on the boundary conditions under which they succeed or falter, and what this implies for design.

4.1 Legibility Conditions for Unplugged Representations

Our comparison suggests that the two toolkits differed in their legibility conditions—the extent to which learners could connect card-level representations to meaningful referents without additional support. For the Deepfake toolkit, everyday media experience provided ready-made referents[2, 8]. For the Agent toolkit, no comparable experiential base existed among rural participants; without grounding, the cards' labels remained opaque—abstract terms that could not be mapped onto familiar phenomena[7].

The Agent toolkit was both more abstract and more hierarchically complex, and its terminology was less familiar. These factors

may have jointly contributed to the observed breakdown. Nevertheless, the observation that the same Agent toolkit supported productive co-design among urban participants with some generative AI experience suggests that the absence of experiential anchors played a meaningful role beyond structural complexity alone. By removing technological scaffolding, the physical simplicity of the unplugged format did not translate into cognitive simplicity; instead, it shifted the interpretive burden onto the learner[6].

4.2 The Role of Experiential Demonstration

The facilitator-initiated software demonstration and its effects suggest that introducing brief experiential encounters with the target AI system before using unplugged representations may support reflective decomposition[3]. This observation also clarifies the distinctive value of unplugged toolkits. Whereas software demonstrations reveal a system's behavioral output—what it does—card-based decomposition externalizes its internal structure into manipulable, debatable components[3, 6]. The cards force explicit causal reasoning and structured reflection (e.g., why does this agent need memory? what happens without role specification?) that fluid software interaction tends to obscure. In this framing, unplugged tools and experiential demos are complementary: the demo provides the experiential anchor, while the cards provide the analytical scaffold.

We frame this as a provisional design heuristic rather than a validated pedagogy: our evidence comes from a single, unplanned intervention, and systematic investigation would be needed to establish generalizability. Nevertheless, the rapid shift from confusion and abandonment to active card retrieval and revision suggests this direction warrants further study.

4.3 Implications for Inclusive AI Education

More broadly, our findings raise a cautionary note for AI literacy initiatives that rely primarily on unplugged approaches in low-resource contexts[11]. If the assumed accessibility of tangible tools does not hold for abstract AI systems, then the gap between learners with and without prior AI experience may persist even when physical access to educational materials is equalized. This is consistent with prior observations that non-experts often default to output-oriented interaction with AI systems when they lack appropriate mental models[13], risking a divide defined not by access to technology but by access to meaning. Addressing this requires attention not only to toolkit design but also to the experiential prerequisites that make those toolkits legible—whether provided through brief software demonstrations, video walkthroughs, or peer mentoring from more experienced colleagues.

4.4 Limitations and Future Work

Several scope limitations should be noted. Our study design cannot fully disentangle the contributions of concept abstractness, toolkit structural complexity, and terminology unfamiliarity to the observed breakdown; these factors co-varied across the two toolkits. The differences between urban and rural workshops may also reflect additional variables beyond experiential familiarity, including group composition, facilitation style, and regional educational norms, which we did not systematically control. Given that the demonstration was a single unplanned intervention, future work

should include controlled comparisons that isolate concept type from toolkit complexity, systematically test whether pairing experiential demonstrations with unplugged toolkits improves legibility across diverse learner populations, and examine how varying forms of scaffolding—such as live demonstrations, video walkthroughs, and guided peer interaction—differentially support the interpretation of abstract AI concepts in unplugged settings.

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